

SAMOS Core 3 – Capstone Group Assignment

Students:

- Duma Amanda (DMXAMA006)
 - Ndoda Ndiphiwe (NDDNDI005)
 - Ranjit Kiara (RNJKIA002)
-

St Helena Bay Chlorophyll and Bathymetry Analysis

Introduction

St Helena Bay is a productive coastal region located along the west coast of South Africa within the Southern Benguela Upwelling System. The region is characterised by strong wind-driven coastal upwelling, which transports nutrient-rich deep waters to the surface and supports high phytoplankton productivity and elevated chlorophyll-a concentrations (Pitcher et al., 2010). The bathymetry of the continental shelf and coastal circulation patterns play an important role in controlling biological productivity and seasonal variability within the region (Lamont et al., 2015). One of the most active of these centres is the Cape Columbine zone adjacent to St Helena Bay (Pitcher, 1988). Upwelling systems are therefore susceptible to harmful algal blooms, as a result to being highly productive nutrient-enriched environments, prone to high-biomass blooms (Pitcher, 1988).

Data Description

1. Bathymetry Data

Description

- **Dataset Name:** GMRTv4_4_1_20260513topo2.grd
- **Source:** Global Multi-Resolution Topography (GMRT) Synthesis, hosted by the Lamont-Doherty Earth Observatory of Columbia University.
- **Spatial Coverage:** High-resolution gridded synthesis. The spatial coverage is a localized regional extract focusing on the South African west coast, spanning latitudes 31.50°S to 33.50°S and longitudes 17.00°E to 18.51°E

- **Temporal Coverage:** Static elevation compilation.
- **Abstract:** The GMRT synthesis blends high-resolution shipborne multibeam sonar data collected by the global scientific community with regional and global basemaps (such as GEBCO) to deliver a continuous, seamless topography and bathymetry model. This regional extract provides the baseline seafloor bathymetry and coastal terrain topography required to contextualize local marine habitats and oceanographic features.

NetCDF Metadata Listing

Dimensions:

- lat: 542
- lon: 343

Coordinates:

- lat(lat):
 - a) type = float64
 - b) range = -33.50 to -31.50 (degrees_north)
- lon(lon):
 - a) type = float64
 - b) range = 17.00 to 18.51 (degrees_east)

Data Variables:

- altitude (lat, lon):
 - a) type = float64
 - b) long_name = "height"
 - c) units = "meters"
 - d) comment = "Elevation and bathymetry values relative to sea level. Negative values indicate seafloor depth below sea level, and positive values indicate terrestrial terrain altitude."

Global Attributes:

- title = "GMRT Grid "
- history = "Projection: Cylindrical Equidistant. Extracted from the Global Multi-Resolution Topography Synthesis."
- Conventions = "COARDS, CF-1.6"
- GMT_version = "4.5.7"

2. Chlorophyll-a Data

Description

- **Dataset Name:** ESACCI-OC-MAPPED-CLIMATOLOGY-1M-MONTHLY_4km_PML_CHL-fv5.0.nc
- **Source:** European Space Agency (ESA) Climate Change Initiative (CCI) Ocean Colour Project, hosted by Plymouth Marine Laboratory (PML). Original satellite data sourced from NASA SeaWiFS (L1A and L2 R2018.0 LAC).
- **Spatial Coverage:** 4 km spatial resolution (0.04 grid intervals). The spatial coverage is subsetting regionally to the South African coastline, spanning latitudes 31.52°S to 33.48°S and longitudes 17.02°E to 18.48°E.
- **Temporal Coverage:** 1-Month Monthly Climatology. The baseline multi-decadal time series spans from January 1998 through to January 2020.
- **Abstract:** This dataset provides a regional extract of the ESA CCI Ocean Colour multi-sensor hybrid product, focusing on near-surface chlorophyll-a concentrations (mg/m^{-3}). The climatology blends long-term satellite observations to characterize seasonal cycles and baseline phytoplankton biomass variations within local coastal upwelling systems.

NetCDF Metadata Listing

Dimensions:

- time = 12
- lat = 48
- lon = 36

Coordinates:

- time(time):
 - a) type = datetime64[ns]
 - b) range = 1998-01-01 to 1997-12-01 (12 climatological months)
- lat(lat):
 - a) type = float64
 - b) range = -31.52 to -33.48 (degrees_north)
- lon(lon):
 - a) type = float64
 - b) range = 17.02 to 18.48 (degrees_east)

Data Variables:

- crs:
 - a) type = int32
 - b) description = Coordinate Reference System / spatial projection parameters
- chlor_a(time, lat, lon):
 - a) type = float32
 - b) long_name = "Chlorophyll-a concentration in sea water using a multi-sensor hybrid algorithm"
 - c) standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water"
 - d) units = "mg/m-3"

Global Attributes:

- title = "ESA CCI Ocean Colour Product (CCI ALL-v5.0-MONTHLY CLIMATOLOGY)"
- institution = "Plymouth Marine Laboratory"
- Conventions = "CF-1.7"
- Metadata_Conventions = "Unidata Dataset Discovery v1.0"
- CDI = "Climate Data Interface"
- CDO = "Climate Data Operators version 1.9.3"
- NCO = "4.7.2"
- history = "Subsetted via Climate Data Operators (cdo selvar,chlor_a)"

Figures

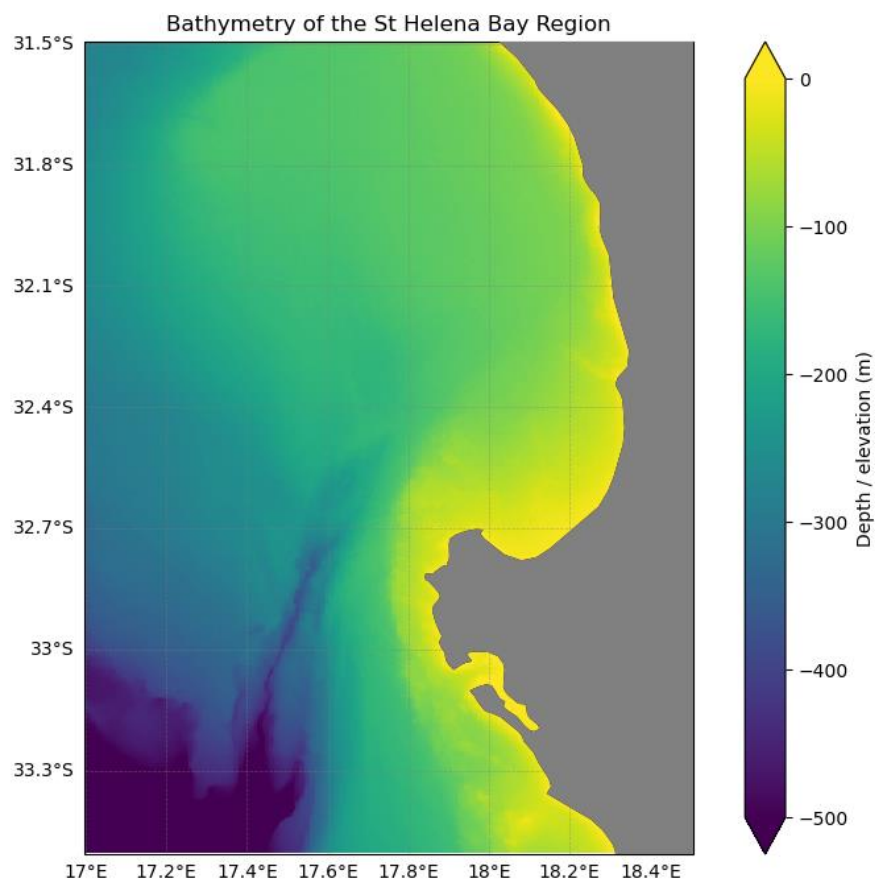


Figure 1: Bathymetry of the St Helena Bay region. Dark blue and purple colours represent deeper offshore waters, while green and yellow colours indicate shallower regions along the continental shelf and coastal zone.

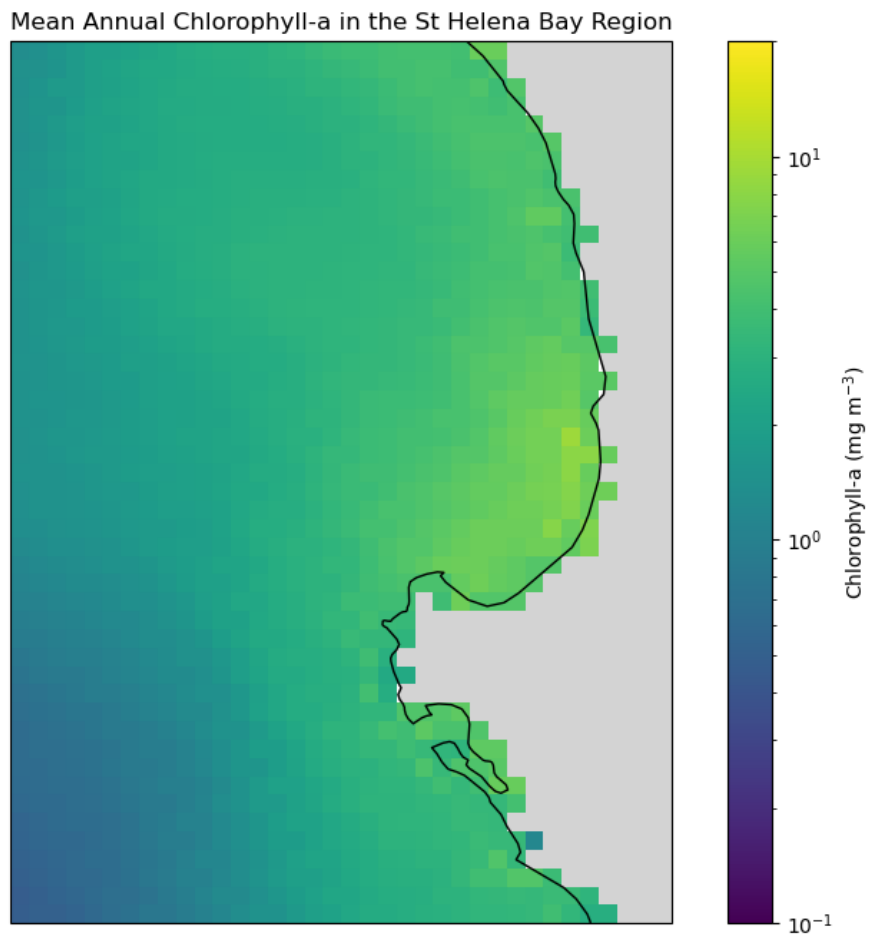


Figure 2: Mean annual chlorophyll-a concentrations in the St Helena Bay region. Higher chlorophyll-a concentrations are represented by yellow colours, while lower concentrations are shown in blue and purple shades

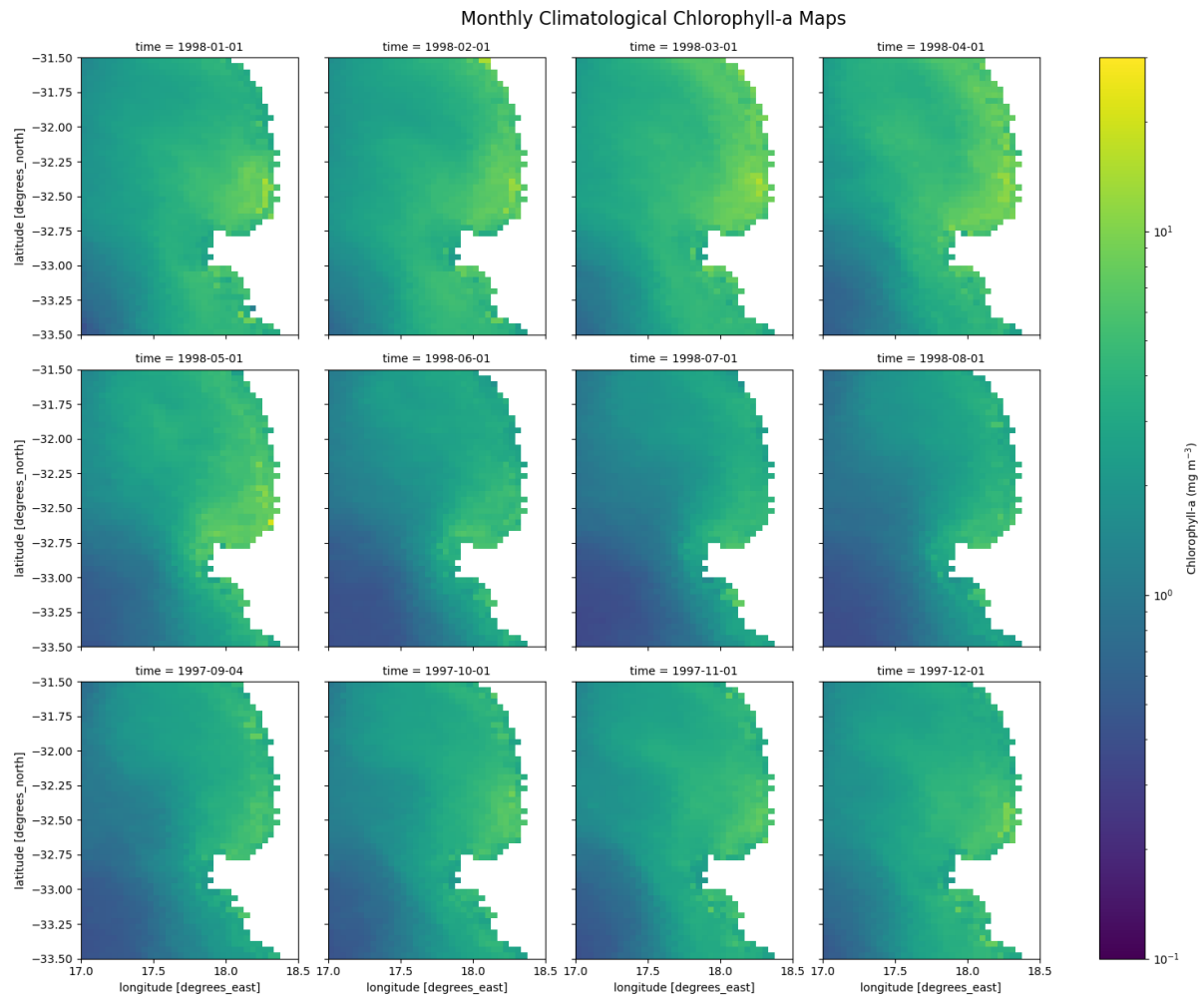


Figure 3: Monthly climatological chlorophyll-a maps for the St Helena Bay region showing seasonal variability in phytoplankton biomass throughout the year 1998.

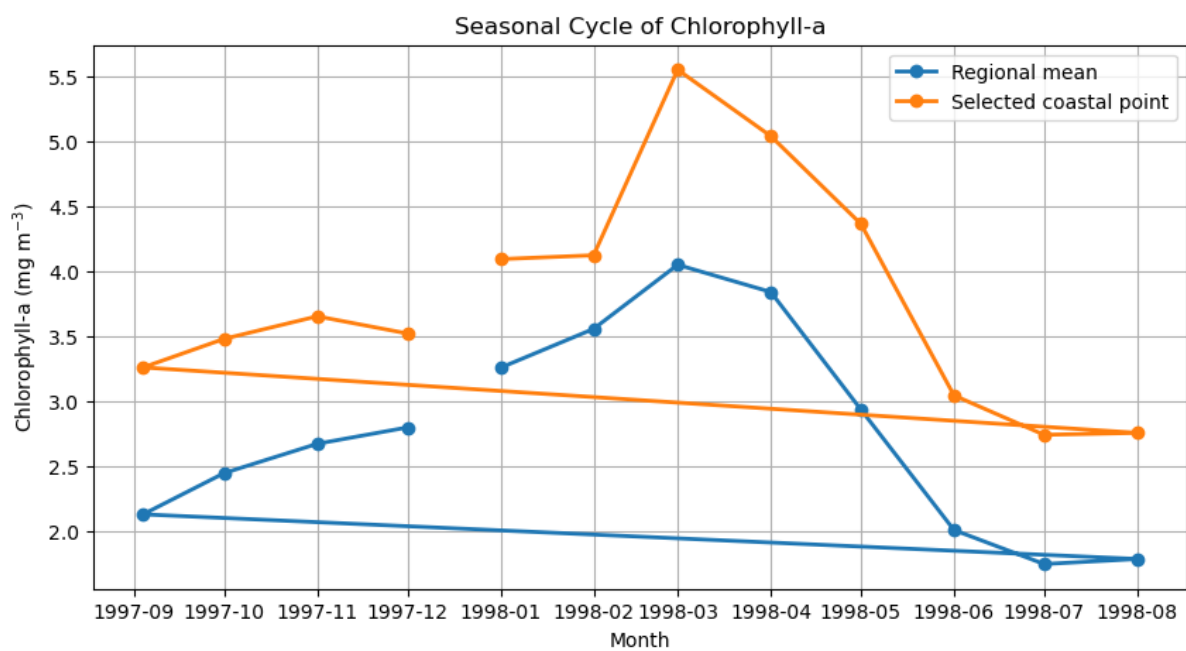


Figure 4: Time series comparing the regional mean seasonal cycle of Chlorophyll-a (mg/m³) against a selected high-concentration coastal grid point from September 1997 to August 1998.

Disclaimer

ChatGPT and Copilot were used to fix the codes for the figures produced.

Gemini was used to ask for a data description structure.

Prompt examples:

- “Suggest colour maps suitable for plotting chlorophyll concentration in Python, considering logarithmic scaling and oceanographic conventions”
- “Check for any errors in this code...”
- “Please show me a template of how to provide a description of a given dataset along with its metadata listings, use the basic structure of a netcdf data file as an example”
- “Show me Python code to save multiple matplotlib figures (bathymetry map, annual chlorophyll map, monthly faceted maps, and seasonal cycle time series) as high-resolution PNG files”

References

1. Lamont, T., Barlow, R.G. and Kyewalyanga, M.S., 2015. Hydrographic variability in the St Helena Bay region of the southern Benguela upwelling system. *Journal of Geophysical Research: Oceans*, 120(4), pp.2920-2939.
2. Pitcher, G.C., 1988. Mesoscale heterogeneities of the phytoplankton distribution in St Helena Bay, South Africa, following an upwelling event. *South African Journal of Marine Science*, 7(1), pp.9-23.
3. Pitcher, G.C., Figueiras, F.G., Hickey, B.M. and Moita, M.T., 2010. The physical oceanography of upwelling systems and the development of harmful algal blooms. *Progress in oceanography*, 85(1-2), pp.5-32.
4. ESA Ocean Colour Climate Change Initiative (OC-CCI) dataset.
5. Global Multi-Resolution Topography (GMRT) Synthesis dataset.

